SULIT
EC015
*Chemistry 1*
*Semester I*
*Session 2020/2021*
*2 hours*

EC015
Kimia 1
Semester I
Sesi 2020/2021
2 jam

**KEMENTERIAN PENDIDIKAN MALAYSIA**

**BAHAGIAN MATRIKULASI**
*MATRICULATION DIVISION*

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**PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI**
*MATRICULATION PROGRAMME EXAMINATION*

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**JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.**
*DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.*

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**INSTRUCTIONS TO CANDIDATE:**

This question paper consists of 6 questions.

Answer **all** questions.

Answers must be written in the answer booklet provided.

All steps must be shown clearly. Use a new page for each question.

Maximum marks awarded are shown in brackets at the end of each question or section.

The use of non-programmable scientific calculator is permitted.

***ARAHAN KEPADA CALON*:**

*Kertas soalan ini mengandungi* **6** *soalan.*

*Jawab* ***semua*** *soalan.*

*Jawapan hendaklah ditulis pada buku jawapan yang disediakan.*

*Semua langkah kerja hendaklah ditunjukkan dengan jelas. Gunakan muka surat baharu bagi nombor soalan yang berbeza.*

*Markah maksimum yang diperuntukkan ditunjukkan dalam kurungan pada hujung setiap soalan atau bahagian soalan.*

*Penggunaan kalkulator saintifik yang tidak boleh diprogramkan dibenarkan.*



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## RELATIVE ATOMIC MASSES OF SELECTED ELEMENTS
## *JISIM ATOM RELATIF UNSUR-UNSUR TERPILIH*

| Element | Symbol | Proton Number | Relative Atomic Mass |
|---|---|---|---|
| Aluminium | Al | 13 | 27.0 |
| Silver | Ag | 47 | 107.9 |
| Argon | Ar | 18 | 40.0 |
| Arsenic | As | 33 | 74.9 |
| Gold | Au | 79 | 197.0 |
| Barium | Ba | 56 | 137.3 |
| Beryllium | Be | 4 | 9.0 |
| Bismuth | Bi | 83 | 209.0 |
| Boron | B | 5 | 10.8 |
| Bromine | Br | 35 | 79.9 |
| Iron | Fe | 26 | 55.9 |
| Fluorine | F | 9 | 19.0 |
| Phosphorus | P | 15 | 31.0 |
| Helium | He | 2 | 4.0 |
| Mercury | Hg | 80 | 200.6 |
| Hydrogen | H | 1 | 1.0 |
| Iodine | I | 53 | 126.9 |
| Cadmium | Cd | 48 | 112.4 |
| Potassium | K | 19 | 39.1 |
| Calcium | Ca | 20 | 40.1 |
| Carbon | C | 6 | 12.0 |
| Chlorine | Cl | 17 | 35.5 |
| Cobalt | Co | 27 | 58.9 |
| Krypton | Kr | 36 | 83.8 |
| Chromium | Cr | 24 | 52.0 |
| Copper | Cu | 29 | 63.6 |
| Lithium | Li | 3 | 6.9 |
| Magnesium | Mg | 12 | 24.3 |
| Manganese | Mn | 25 | 54.9 |
| Sodium | Na | 11 | 23.0 |
| Neon | Ne | 10 | 20.2 |
| Nickel | Ni | 28 | 58.7 |
| Nitrogen | N | 7 | 14.0 |
| Oxygen | O | 8 | 16.0 |
| Platinum | Pt | 78 | 195.1 |
| Lead | Pb | 82 | 207.2 |
| Protactinium | Pa | 91 | 231.0 |
| Radium | Ra | 88 | 226.0 |
| Radon | Rn | 86 | 222.0 |
| Rubidium | Rb | 37 | 85.5 |
| Selenium | Se | 34 | 79.0 |
| Cerium | Ce | 58 | 140.1 |
| Cesium | Cs | 55 | 132.9 |
| Silicon | Si | 14 | 28.1 |
| Scandium | Sc | 21 | 45.0 |
| Tin | Sn | 50 | 118.7 |
| Antimony | Sb | 51 | 121.8 |
| Strontium | Sr | 38 | 87.6 |
| Sulphur | S | 16 | 32.1 |
| Uranium | U | 92 | 238.0 |
| Tungsten | W | 74 | 183.9 |
| Zinc | Zn | 30 | 65.4 |



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## LIST OF SELECTED CONSTANT VALUES
## *SENARAI NILAI PEMALAR TERPILIH*

| | | | |
|---|---|---|---|
| Ionisation constant for water at 25 °C<br>*Hasil darab ion bagi air pada* 25 °C | $K_w$ | = | $1.00 \times 10^{-14}$ mol$^2$ dm$^{-6}$ |
| Molar volume of gases<br>*Isi padu molar gas* | $V_m$ | =<br>= | 22.4 dm$^3$ mol$^{-1}$ at STP<br>24 dm$^3$ mol$^{-1}$ at room temperature |
| Speed of light in a vacuum<br>*Laju cahaya dalam vakum* | $c$ | = | $3.0 \times 10^8$ m s$^{-1}$ |
| Specific heat of water<br>*Muatan haba tentu air* | | =<br>=<br>= | 4.18 kJ kg$^{-1}$ K$^{-1}$<br>4.18 J g$^{-1}$ K$^{-1}$<br>4.18 J g$^{-1}$ °C$^{-1}$ |
| Avogadro's number<br>*Nombor Avogadro* | $N_A$ | = | $6.02 \times 10^{23}$ mol$^{-1}$ |
| Faraday constant<br>*Pemalar Faraday* | $F$ | = | $9.65 \times 10^4$ C mol$^{-1}$ |
| Planck's constant<br>*Pemalar Planck* | $h$ | = | $6.6256 \times 10^{-34}$ J s |
| Rydberg constant<br>*Pemalar Rydberg* | $R_H$ | =<br>= | $1.097 \times 10^7$ m$^{-1}$<br>$2.18 \times 10^{-18}$ J |
| Molar of gases constant<br>*Pemalar gas molar* | $R$ | =<br>= | 8.314 J K$^{-1}$ mol$^{-1}$<br>0.08206 L atm mol$^{-1}$ K$^{-1}$ |
| Density of water<br>*Ketumpatan air* | $\rho$ | = | 1 g cm$^{-3}$ |
| Freezing point of water<br>*Takat beku air* | | = | 0.00°C |
| Vapour pressure of water<br>*Tekanan wap air* | $P_{water}$ | = | 23.8 torr |

## UNIT AND CONVERSION FACTOR
## *UNIT DAN FAKTOR PERTUKARAN*

| | |
|---|---|
| Volume<br>*Isi padu* | 1 liter = 1 dm$^3$<br>1 mL = 1 cm$^3$ |
| Energy<br>*Tenaga* | 1 J = 1 kg m$^2$ s$^{-2}$ = 1 N m = $10^7$ erg<br>1 calorie = 4.184 Joule<br>1 eV = $1.602 \times 10^{-19}$ J |
| Pressure<br>*Tekanan* | 1 atm = 760 mm Hg = 760 torr = 101.325 kPa = 101325 N m$^{-2}$ |
| Others<br>*Lain-lain* | 1 Faraday(F) = 96500 coulomb<br>1 Newton(N) = 1 kg m s$^{-2}$ |



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Answer **all** questions.
*Jawab **semua** soalan.*

**1** (a) Sodium metasilicate, $Na_2SiO_3$, is used in the production of silica gel.
*Natrium metasilikat,* $Na_2SiO_3$, *digunakan dalam pengeluaran gel silika.*

(i) Write the isotopic notation for silicon atom.
*Tulis notasi isotop untuk atom silikon.*

(ii) Calculate the number of Si atoms in 50 g of $Na_2SiO_3$.
*Kirakan bilangan atom* Si *dalam* 50 g $Na_2SiO_3$.

[3 *marks*]
[3 *markah*]

(b) The following reaction takes place in an acidic condition.
*Tindak balas berikut berlaku dalam keadaan berasid.*

$$MnO_4^- + Fe^{2+} \rightarrow Mn^{2+} + Fe^{3+}$$

Write a balance equation for the above reaction.
*Tulis suatu persamaan berimbang untuk tindak balas tersebut.*

[3 *marks*]
[3 *markah*]

(c) Compound **X** has a molar mass of 294.20 g. Analysis of **X** shows that it contains 26.58 g of potassium, 35.35 g of chromium and 38.07 g of oxygen. Determine the molecular formula and name **X**.
*Sebatian* **X** *mempunyai jisim molar* 294.20 g. *Analisis* **X** *menunjukkan bahawa ia mengandungi* 26.58 g *kalium*, 35.35 g *kromium dan* 38.07 g *oksigen. Tentukan formula molekul dan namakan* **X**.

[6 *marks*]
[6 *markah*]



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(d) The reaction between nitrogen dioxide, $NO_2$, with water produces nitric acid, $HNO_3$, and nitrogen monoxide, NO. In a reaction between 100.0 g of $NO_2$ and 50.0 g of water, 80.0 g of $HNO_3$ is produced.

*Tindak balas antara nitrogen dioksida,* $NO_2$, *dengan air menghasilkan asid nitrik,* $HNO_3$, *dan nitrogen monoksida,* NO. *Dalam suatu tindak balas antara* 100.0 g $NO_2$ *dan* 50.0 g *air,* 80.0 g $HNO_3$ *telah dihasilkan.*

(i) Determine the limiting reactant in the reaction.

*Tentukan reaktan penghad di dalam tindak balas.*

(ii) Determine the percentage yield of the reaction.

*Tentukan peratusan hasil tindak balas.*

[9 *marks*]
[9 *markah*]

2 (a) Calculate the energy emitted and the wavelength of the spectrum line when an electron moves from $n = 6$ to $n = 3$. State the series and electromagnetic region of the spectrum line.

*Kira tenaga yang dipancarkan dan panjang gelombang garis spektrum apabila elektron bergerak dari* $n = 6$ *ke* $n = 3$. *Nyatakan siri dan kawasan elektromagnet bagi garis spektrum ini.*

[6 *marks*]
[6 *markah*]

(b) An electron has the following quantum numbers: $n = 3$, $\ell = 2$. Identify its orbital. Draw the spatial orientation for all the possible orbitals.

*Suatu elektron mempunyai nombor kuantum berikut:* $n = 3$, $\ell = 2$. *Kenal pasti orbitalnya. Lukis orientasi ruang bagi kesemua orbital yang mungkin.*

[4 *marks*]
[4 *markah*]



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3 Phosgene, $COCl_2$, is a chemical used in the production of plastics and pesticides. Given that chlorine cannot be the central atom,

*Fosgena,* $COCl_2$, *adalah bahan kimia yang digunakan dalam pengeluaran plastik dan racun perosak. Diberi klorin tidak boleh menjadi atom pusat,*

(a) draw **three (3)** possible structures of phosgene.

*lukis* ***tiga*** **(3)** *struktur fosgena yang mungkin.*

[3 *marks*]
[3 *markah*]

(b) determine the most plausible structure and give your reason.

*tentukan struktur yang paling munasabah dan berikan alasan anda.*

[5 *marks*]
[5 *markah*]

(c) determine the hybrid orbital for the central atom of the most plausible structure.

*tentukan orbital hibrid atom pusat bagi struktur yang paling munasabah.*

[4 *marks*]
[4 *markah*]

(d) name the molecular shape and draw the overlapping orbitals of $COCl_2$.

*namakan bentuk molekul dan lukis pertindihan orbital bagi* $COCl_2$.

[5 *marks*]
[5 *markah*]



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4 (a) Sketch a labelled phase diagram of carbon dioxide.

*Lakarkan suatu gambar rajah fasa karbon dioksida berlabel.*

[4 *marks*]
[4 *markah*]

(b) 2.0 g of He and 61 g of $O_2$ were placed in a 5.0-L tank at 25 °C. Determine the partial pressure and the total pressure of the gas mixture.

2.0 g He *dan* 61 g $O_2$ *diletakkan dalam tangki* 5.0-L *pada* 25 °C. *Tentukan tekanan separa dan jumlah tekanan campuran gas tersebut.*

[5 *marks*]
[5 *markah*]

5 (a) Ammonia, $NH_3$, is produced at 500 °C and 200 atm in the presence of $Fe_2O_3$ catalyst according to the following equation:

*Ammonia*, $NH_3$, *dihasilkan pada* 500 °C *dan* 200 atm *dengan kehadiran mangkin* $Fe_2O_3$ *mengikut persamaan berikut*:

$$3H_2(g) + N_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92.22 \text{ kJ}$$

Suggest **two (2)** changes that can be made to increase the yield of $NH_3$. Explain your suggestions.

*Cadangkan* ***dua*** **(2)** *perubahan yang boleh dibuat untuk meningkatkan hasil* $NH_3$. *Terangkan cadangan anda.*

[4 *marks*]
[4 *markah*]

(b) Consider the following reaction:

*Pertimbangkan tindak balas berikut*:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \qquad K_c = 4.62$$

Determine the direction of the reaction if the concentrations of $SO_2(g)$, $SO_3(g)$ and $O_2(g)$ are 0.2 M, 0.15 M and $8.55 \times 10^{-3}$ M, respectively.

*Tentukan arah tindak balas jika kepekatan* $SO_2(g)$, $SO_3(g)$ *dan* $O_2(g)$ *masing-masing ialah* 0.2 M, 0.15 M *dan* $8.55 \times 10^{-3}$ M.

[5 *marks*]
[5 *markah*]



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6 (a) Determine the pH of 0.5 M ethanoic acid, $CH_3COOH$, solution.
[Given: $K_a$ $CH_3COOH = 1.8 \times 10^{-5}$]

*Tentukan* pH *larutan asid etanoik*, $CH_3COOH$, 0.5 M.
[*Diberi*: $K_a$ $CH_3COOH = 1.8 \times 10^{-5}$]

[5 *marks*]
[5 *markah*]

(b) In an experiment, 25 mL of 0.5 M $CH_3COOH$ was titrated against 0.5 M NaOH solution using phenolphthalein as indicator. The end point is reached when 25 mL of NaOH solution was added.

*Dalam suatu eksperimen*, 25 mL 0.5 M $CH_3COOH$ *dititrat dengan larutan* 0.5 M NaOH *menggunakan fenolftalein sebagai penunjuk. Titik akhir dicapai apabila* 25 mL *larutan* NaOH *ditambah.*

(i) Determine the concentration of the product.

*Tentukan kepekatan produk.*

(ii) Determine the pH of the solution at the end point.

*Tentukan* pH *larutan pada titik akhir.*

[9 *marks*]
[9 *markah*]

**END OF QUESTION PAPER**
***KERTAS SOALAN TAMAT***



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